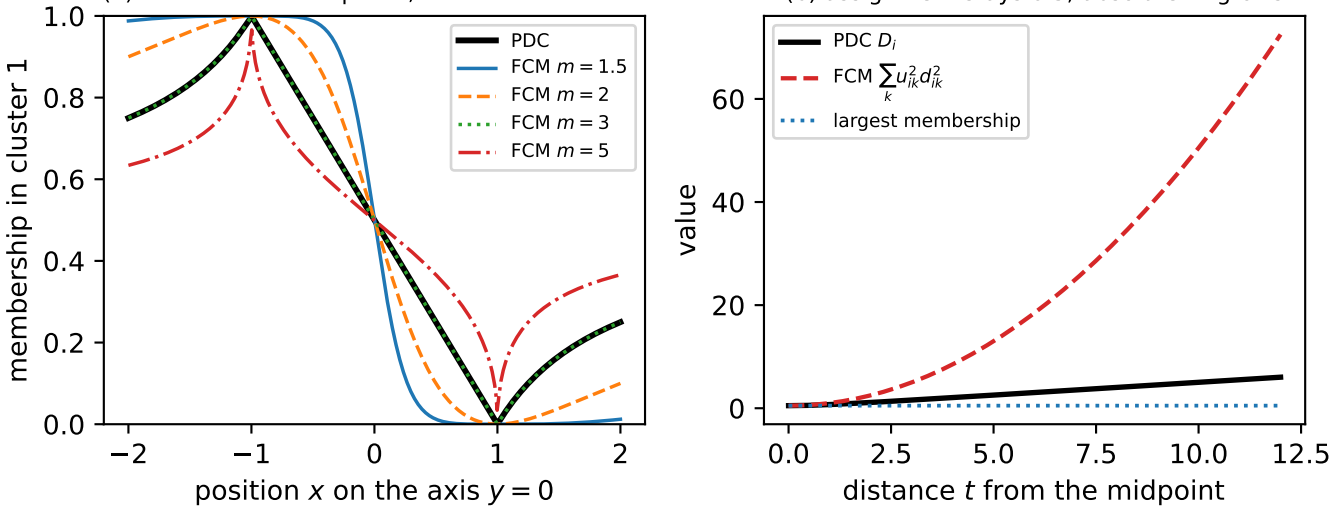
(a) fuzzifier sets sharpness; $m = 3$ matches PDC
membership in cluster 1
position x on the axis $y = 0$
PDC
FCM $m = 1.5$
FCM $m = 2$
FCM $m = 3$
FCM $m = 5$
(b) assignment stays 0.5; absolute fit grows
value
distance t from the midpoint
PDC D_i
FCM $\sum_k u_{ik}^2 d_{ik}^2$
largest membership